Assignment: Implementing a Business Solution with Deep Learning

Business Value:

The proposed deep learning model for hotel cancellation prediction provides significant business value to the hospitality industry by enabling hotels to anticipate customer behaviour more accurately. Since last-minute cancellations directly affect occupancy rates, revenue generation, staffing, and inventory planning, the model helps hotels make proactive business decisions. By identifying high-risk bookings in advance, hotel management can implement targeted strategies such as personalized customer engagement, flexible pricing, promotional offers, or controlled overbooking to minimize financial losses caused by cancellations.

From a practical perspective, the model can be integrated into existing hotel reservation and customer relationship management systems with minimal operational disruption. The prediction results can support departments such as revenue management, operations, and marketing in improving forecasting accuracy and resource allocation. For example, hotels can optimize room pricing based on expected occupancy levels, reduce unnecessary operational costs, and improve workforce scheduling during peak and low-demand periods. This makes the solution highly relevant for modern data-driven hospitality businesses.

The potential impact of this project extends beyond operational efficiency. Accurate cancellation prediction can improve customer satisfaction by enabling more personalized communication and better service planning. Additionally, hotels can use predictive insights to maintain stable occupancy rates and maximize profitability in competitive markets. Overall, the deep learning model demonstrates how artificial intelligence can transform traditional hospitality operations into smarter, more efficient, and strategically informed business processes.

Technical Implementation:

Selected the dataset from Kaggle. (/kaggle/input/dl-course-data/hotel.csv)

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Pre-processed the data using Python functions

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Split the data

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Defined the data (the model has both batch normalization and dropout layers)

Model contains 2 hidden layers, 1 output layer with sigmoid activation function

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Added Optimizer, Loss and Metric

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Trained (200 epochs) and Evaluated the model (Epoch no. 23/200 was recorded with highest accuracy:0.83 and lowest entropy:0.37)

Conclusion:

It can be seen as the training loss continuing to fall, the early stopping callback prevented any over-fitting. Moreover, the accuracy rose at the same rate as the cross-entropy fell, so it appears that minimizing cross-entropy was a good stand-in. All in all, it looks like this training was a success!

Google Colab Link:

https://colab.research.google.com/drive/1hKJSPko8fcUxgSozOUbiwDEp5FTn-VHS?usp=sharing